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CSES Problem Set

Inverse Inversions

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to create a permutation of numbers $1, 2, \dots, n$ that has exactly k inversions.

An inversion is a pair (a, b) where $a < b$ and $p_a > p_b$ where p_i denotes the number at position i in the permutation.

Input

The only input line has two integers n and k .

Output

Print a line that contains the permutation. You can print any valid solution.

Constraints

- $1 \leq n \leq 10^6$
- $0 \leq k \leq \frac{n(n-1)}{2}$

Example

Input:

5 4

Output:

1 5 2 4 3

CSES Problem Set

Monotone Subsequences

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to create a permutation of numbers $1, 2, \dots, n$ whose longest monotone subsequence has exactly k elements.

A monotone subsequence is either increasing or decreasing. For example, some monotone subsequences in $[2, 1, 4, 5, 3]$ are $[2, 4, 5]$ and $[4, 3]$.

Input

The first input line has an integer t : the number of tests.

After this, there are t lines. Each line has two integers n and k .

Output

For each test, print a line that contains the permutation. You can print any valid solution. If there are no solutions, print IMPOSSIBLE.

Constraints

- $1 \leq t \leq 1000$
- $1 \leq k \leq n \leq 100$

Example

Input:

```
3
5 3
5 2
7 7
```

Output:

```
2 1 4 5 3
IMPOSSIBLE
1 2 3 4 5 6 7
```

CSES Problem Set**Chess Tournament**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There will be a chess tournament of n players. Each player has announced the number of games they want to play.

Each pair of players can play at most one game. Your task is to determine which games will be played so that everybody will be happy.

Input

The first input line has an integer n : the number of players. The players are numbered $1, 2, \dots, n$.

The next line has n integers $x_1, x_2, \dots, x_n$: for each player, the number of games they want to play.

Output

First print an integer k : the number of games. Then, print k lines describing the games. You can print any valid solution.

If there are no solutions, print "IMPOSSIBLE".

Constraints

- $1 \leq n \leq 10^5$
- $\sum_{i=1}^n x_i \leq 2 \cdot 10^5$

Example

Input:

```
5
1 3 2 0 2
```

Output:

```
4
1 2
2 3
2 5
3 5
```